
Targeted Lobbying on Council Networks

Final Presentation

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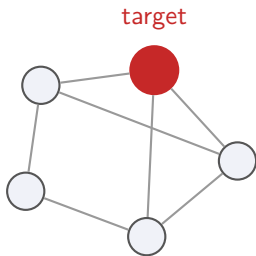
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Whom should a lobbyist target?

A lobbyist with a limited budget.

A council connected by party, committee, and personal ties.

Pressure on one member spills over to others.



What we keep from the literature

Shleifer & Vishny (1993). Bribes shift incentives. We keep this as the lobbyist effort t_i in the payoff.

Galeotti, Golub & Goyal (2020). Linear-quadratic network game; planner intervenes to maximize welfare.

Our move. Swap welfare-maximization for *minimum cost to pass a bill*. The problem becomes a linear program.

Old answer: eigenvector centrality.

New answer: Katz–Bonacich centrality with decay β/c .

The model

$$U_i = a_i \left(b_i + t_i + \beta \sum_j g_{ij} a_j \right) - \frac{1}{2} c a_i^2$$

b_i : baseline inclination toward the bill

t_i : lobbyist effort placed on member i

β, c : peer-effect strength and own-action cost

Lobbyist commits to $t \geq 0$, council picks a , bill passes if $\sum_i a_i \geq \bar{T}$.

Solving for the equilibrium: best response

Each U_i is strictly concave in a_i (since $c > 0$), so the first-order condition is sufficient. Differentiate U_i with respect to a_i :

$$\frac{\partial U_i}{\partial a_i} = b_i + t_i + \beta \sum_{j \in N} g_{ij} a_j - c a_i = 0.$$

Solve for a_i to get member i 's best response, given everyone else's actions:

$$a_i = \frac{1}{c} \left(b_i + t_i + \beta \sum_{j \in N} g_{ij} a_j \right).$$

Each member responds linearly to three things: their own preference, the lobbyist's push, and a peer-weighted average of neighbors' actions.

Solving for the equilibrium

Stack the n best responses into vector form:

$$c \mathbf{a} = (\mathbf{b} + \mathbf{t}) + \beta G \mathbf{a} \iff (cI - \beta G) \mathbf{a} = \mathbf{b} + \mathbf{t}.$$

Under the spectral condition $c > \beta \lambda_{\max}(G)$, the matrix $cI - \beta G$ is positive definite, so it inverts cleanly:

$$\boxed{a^*(t) = M(b + t), \quad M := (cI - \beta G)^{-1}}$$

Strict concavity of each U_i in a_i makes this the unique Nash equilibrium of the council subgame.

Council equilibrium

Under the spectral-radius condition $c > \beta \lambda_{\max}(G)$, the council has a unique Nash equilibrium:

$$a^*(t) = M(b + t), \quad M := (cI - \beta G)^{-1}.$$

Katz–Bonacich centrality with decay β/c :

$$v := M\mathbf{1}.$$

v_i is the aggregate council support generated when the lobbyist places one unit of effort on member i and zero on everyone else.

The lobbyist's problem becomes an LP

Aggregate support is linear in t :

$$\sum_i a_i^*(t) = v^\top b + v^\top t.$$

$$\min_{t \geq 0} \mathbf{1}^\top t \quad \text{s.t.} \quad v^\top t \geq \Delta := \bar{T} - v^\top b$$

The whole game collapses to one linear constraint on the non-negative orthant.

Theorem 1 — Concentration

Theorem 1 (Concentration on max Katz–Bonacich).

Suppose $\Delta = \bar{T} - v^\top b > 0$ and let $i^* \in \arg \max_i v_i$. A cost-minimizing plan loads the entire budget on i^* , with

$$t_{i^*}^* = \frac{\Delta}{v_{i^*}}, \quad t_j^* = 0 \quad \forall j \neq i^*,$$

and minimum cost

$$C^*(\bar{T}) = \frac{\bar{T} - v^\top b}{\max_i v_i}.$$

The lobbyist puts the *whole* budget on the member with the highest Katz–Bonacich centrality. No spreading.

Proof of Theorem 1: lower bound on cost

Take any feasible $t \geq 0$ with $v^\top t \geq \Delta$. The constraint chains into a lower bound on the total spend:

$$\Delta \leq v^\top t = \sum_{i \in N} v_i t_i \leq \max_i v_i \cdot \sum_i t_i = v_{i^*} \cdot \mathbf{1}^\top t.$$

Divide through by v_{i^*} :

$$\boxed{\mathbf{1}^\top t \geq \frac{\Delta}{v_{i^*}}}$$

No feasible plan can achieve cost below Δ divided by the maximum Katz–Bonacich centrality.

Proof of Theorem 1

We now exhibit a feasible plan that hits the lower bound. Define $\hat{t}$ to put the entire budget on member i^* :

$$\hat{t} := \frac{\Delta}{v_{i^*}} e_{i^*}, \quad \text{zero on every other member.}$$

Three quick checks:

- **Non-negative.** $\hat{t} \geq 0$. ✓
- **Feasible.** $v^\top \hat{t} = v_{i^*} \cdot \frac{\Delta}{v_{i^*}} = \Delta$. ✓
- **Cost.** $\mathbf{1}^\top \hat{t} = \frac{\Delta}{v_{i^*}}$ — matches the lower bound. ✓

The all-on- i^* corner achieves the floor, so it is optimal. ■

A worked example: 3-member path



Parameters

$$c = 1, \quad \beta = 0.4$$

Baseline beliefs $b = \mathbf{0}$

Passage threshold $\bar{T} = 1$

Spectral check: $\beta \lambda_{\max}(G) = 0.4\sqrt{2} \approx 0.57 < 1 = c$. ✓

Adjacency matrix

$$G = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

A worked example

With $c = 1$ and $\beta = 0.4$: $\det(cI - \beta G) = 1 - 2\beta^2 = 0.68$. Invert:

$$M = (cI - \beta G)^{-1} \approx \begin{pmatrix} 1.235 & 0.588 & 0.235 \\ 0.588 & 1.471 & 0.588 \\ 0.235 & 0.588 & 1.235 \end{pmatrix}.$$

Each v_i sums column i of M — the total response to lobbying member i :

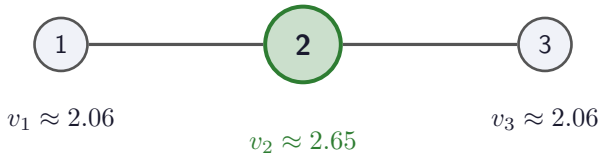
$$v_1 = \underbrace{1.235}_{\text{own}} + \underbrace{0.588}_{\text{node 2}} + \underbrace{0.235}_{\text{node 3}} \approx 2.06$$

$$v_2 = \underbrace{0.588}_{\text{node 1}} + \underbrace{1.471}_{\text{own}} + \underbrace{0.588}_{\text{node 3}} \approx 2.65$$

$$v_3 = \underbrace{0.235}_{\text{node 1}} + \underbrace{0.588}_{\text{node 2}} + \underbrace{1.235}_{\text{own}} \approx 2.06$$

Node 2 wins because two neighbors spill back, which also boosts its own response M_{22} .

Applying Theorem 1



Targeting node 2 costs $1/v_2 \approx 0.378$. Targeting a leaf costs $1/v_1 \approx 0.486$.

Centrality premium: 28.6%.

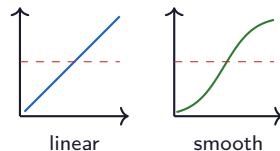
Real voting isn't linear

Real institutions don't aggregate effort linearly.

Replace the linear rule $\sum_i a_i \geq \bar{T}$ with

$$\sum_i f(a_i^*) \geq K$$

for some C^2 strictly increasing f .



Theorem 2 — Slope-weighted Bonacich

Theorem 2 (Slope-weighted concentration).

Let f be C^2 with $f' > 0$, and let $a^0 = Mb$ be the no-lobbying equilibrium. Define the *slope-weighted centrality*

$$w_j = \sum_i f'(a_i^0) M_{ij}, \quad a^0 = Mb.$$

As the support gap $\Delta = K - F(a^0) \downarrow 0$, the cost-minimizing plan concentrates on $j^* \in \arg \max_j w_j$.

Linear f or $b = 0$: w collapses to v . Theorem 1 recovered.

Concave f + nonzero b : ranking can flip — fence-sitters can beat saturated central partisans.

The saturation flip

$b = 0$: center wins



$$w \approx (0.515, 0.662, 0.515)$$

$b = 2$: leaves win



$$a^0 \approx (4.12, 5.29, 4.12) \text{ (saturated)}$$

$$w \approx (0.0261, 0.0259, 0.0261)$$

When the central member is saturated, peripheral fence-sitters become the cheaper target.

Model B: probabilistic votes

Now $a_i \in (0, 1)$ is the probability member i votes yes. Replace the quadratic cost with binary entropy. Best response is sigmoidal:

$$a_i^* = \sigma\left(\frac{b_i + t_i + \beta \sum_j g_{ij} a_j^*}{\kappa}\right).$$

$$\tilde{v}_j = \underbrace{a_j^*(1 - a_j^*)}_{\text{swing slope}} \cdot \underbrace{\left[\text{modified Bonacich}\right]_j}_{\text{network propagation}}$$

The swing-voter slope $\sigma'(a^*)$ plays exactly the role of $f'(a^0)$ from Theorem 2 — the slope-weighting channel is real.

Open directions

1. Directed networks: in-Bonacich (susceptibility) vs out-Bonacich (influence).
2. Competing lobbyists.
3. Repeated/dynamic game extension

Thank you. Questions?